Quantifying statements

Definition 1. A statement is a sentence or mathematical expression that has a logical value: it is either true or false.

Example.

1. Bernardo
2. Bernardo is Brazilian.
3. Spanish is Brazil's official language.
4. The real part of every nontrivial zero of the Riemann zeta function is $\frac{1}{2}$.

Notation: $P, Q, R, S, \ldots$
For instance, one can declare

$$P : \text{If a positive integer } n \text{ is not prime, then } n = pq \text{ for } p, q \in \mathbb{Z},\ p, q > 1.$$

When statements have dependence on n, it is common to write $P(n)$ to explicitly show the dependence. Note that being written as $P(n)$ doesn't guarantee that $P(n)$ is a statement. For instance, the truth value of

$$Q(n) : f(x) = \frac{1}{x} \text{ is continunous at } x = n$$

depends on the value of n. This is called an *open sentence*.
There isn't really a limit to how many parameters an open sentence can have, for instance,

$$R(f, n) : \text{The function } f \text{ is continuous at } n.$$

However, as we have already seen, there are statements that are clearly true or false but not easily proven:

S : [**Goldbach's Conjecture**]Every even integer greater than 2 is a sum of two prime numbers.

Definition 2. Given two statements P, Q, one can define

$$\begin{aligned} P \wedge Q &= P \text{ and } Q \\ P \vee Q &= P \text{ or } Q \\ \sim P &= \text{ not } P \end{aligned}$$

We say that

- $P \wedge Q$ is true when P and Q are true.
- $P \vee Q$ is true when at least one of P, Q is true.
- $\sim P$ is true when P is false

When operating on statements, it is worthwhile to understand all possible logical outcomes. We do that using truth tables.

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

P	$\sim P$
T	F
F	T

Example. Compute truth tables for the following statements.

1. $\sim (P \wedge Q)$
2. $(\sim P) \vee (\sim Q)$
3. $\sim (P \vee Q)$
4. $(\sim P) \wedge (\sim Q)$

Looking at the first two, and the last two, what do you see?

Definition 3. Two statements R, S are said to be logically equivalent if their outcome in the truth table match line by line. When that happens, we say that $R = S$.

Proposition 1 (DeMorgan's Laws)**.** *The following are logically equivalent.*

1. $\sim (P \wedge Q) = (\sim P) \vee (\sim Q)$
2. $\sim (P \vee Q) = (\sim P) \wedge (\sim Q)$

Definition 4. Let P, Q be statements. We define the conditional statement

$$P \implies Q : \text{If } P\text{, then } Q,$$

which means that "If P is true, then Q is true".

B: You can think of $P \implies Q$ as $P \subseteq Q$. The only way for $P \nsubseteq Q$ is to have an element in P that is true and Q is false.
The truth table is given by

P	Q	$P \implies Q$
T	T	T
T	F	F
F	T	T
F	F	T

There are many ways to read the statement "$P \implies Q$".

1. If P, then Q
2. Q if P
3. Q whenever P
4. Q, provided that P
5. whenever P, then Q
6. P is a sufficient condition for Q
7. For Q, it is sufficient that P
8. Q is a necessary condition for P
9. P only if Q

Example. Compute the truth table for

1. $Q \implies P$
2. $\sim (P \implies Q)$
3. $(\sim P) \wedge Q$
4. $\sim Q \implies \sim P$

and comment on your findings.

One of the findings should be that $P \implies Q \neq Q \implies P$. When both those things happen, we write

Definition 5. $P \iff Q = (P \implies Q) \wedge (Q \implies P)$

A truth table is the simplest way to understand the logical values here, so let's do that.

P	Q	$P \implies Q$	$Q \implies P$	$(P \implies Q) \wedge (Q \implies P)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Again, there many ways to read the statement $P \iff Q$:

1. P if and only if Q
2. P is a necessary and sufficient condition for Q
3. For P it is necessary and sufficient that Q
4. P is equivalent to Q
5. If P, then Q, and conversely.

Example.

1. A is invertible if and only if $\det(A) \neq 0$
2. If $xy = 0$ then $x = y$ or $y = 0$ and conversely.

Example. Compute the following truth tables.

1. $\sim P \iff \sim Q$
2. $\sim (P \wedge Q)$
3. $(P \vee Q) \wedge \sim (P \wedge Q)$

The sky is the limit when computing truth tables. Of course nobody enjoys doing them, but just like Venn Diagrams, it helps providing a different framework to look at statements. For instance, you can write a table for three statements P, Q, R as in

P	Q	R	$\sim R$	$Q \wedge \sim R$	$P \vee (Q \wedge \sim R)$
T	T	T	F	F	T
T	T	F	T	F	T
T	F	T	F	F	T
T	F	F	T	F	T
F	T	T	F	F	F
F	T	F	T	T	T
F	F	T	F	F	F
F	F	F	T	F	F

Here is a list of logical equivalences that can be proven either in class or in the bathroom:

Example.
1. DeMorgan's Laws (which should be done by now)
2. Contrapositive: $P \implies Q = (\sim Q \implies \sim P)$
3. Commutative
4. Distributive
5. Associative
6. Example counter-intuitive: $P \vee (Q \wedge R) \neq (P \vee Q) \wedge R$

In general, you can work through a line of the truth table to infer the logical value of statements.

Example. Suppose P is false and $(R \implies S) \iff (P \wedge Q)$ is true. What is the truth value of R, S?
Of course, note that if P is false, then $P \wedge Q$ is false.
Since $(R \implies S) \iff (P \wedge Q)$ is true and $P \wedge Q$ is false, it follows that $R \implies S$ shares the same truth value: false.
Note that $R \implies S$ is only false when R is true and S is false.

Quantifying statements

A few statements that are very common in your textbook (and that I might use are):

1. $E(n)$: n is even
2. $O(n)$: n is odd
3. $P(n)$: n is prime

Definition 6. The symbols $\forall$ and $\exists$ are called quantifiers.
$\forall$ stands for the phrase "for all", "for every", or "for each". It is a universal quantifier.
$\exists$ stands for the phrase "there exists a", "there is a". It is a existence quantifier.

Example. Let us rewrite a few sentences from English to symbolic language:

1. For every non-negative integer n, the sum of n with its successor is odd. $\forall n \in \mathbb{N}, O(n + (n+1))$.
2. There exists a subset of the natural numbers with size equal to 2023. $\exists X \subseteq \mathbb{N}, |X| = 2023$.
3. There exists an subset of the real numbers that it is not the real numbers and it is infinite. $\exists X \subseteq \mathbb{R}, X \neq \mathbb{R} \wedge (|X| > n,\ \forall n \in \mathbb{N}.)$
4. Every integer of the form $4k+1$ for some integer k is prime. $\forall k \in \mathbb{Z}, P(4k+1)$
5. There exists a prime of the form $2^m - 1$. $\exists m \in \mathbb{Z}, P(2^m - 1)$ (Mersenne primes)
6. For every positive integer, $n^2 - n + 41$ is prime. $\forall n \in \mathbb{N}, P(n^2 - n + 41)$.

Proposition 2. *The order matters!*

Example. Explore the difference between the statements

1. $\forall n \in \mathbb{R},\ \exists c \in \mathbb{R},\ x^2 = c$
2. $\exists c \in \mathbb{R}, \forall x \in \mathbb{R}, x^2 = c$

Example. Translate back to English:

1. $\forall n \in \mathbb{N}, \exists X \in \mathcal{P}(\mathbb{N}), |X| < n$
2. $\exists a \in \mathbb{R}, \forall x \in \mathbb{R}, ax = x$
3. $\forall n \in \mathbb{N}, \forall m \in \mathbb{N}, n^m = m^n$
4. $\exists n \in \mathbb{N}, \exists m \in \mathbb{N}, n + m = 2023$

Example. Translate to symbolic:

1. Mean Value Theorem: If f is continuous on $[a, b]$ and differentiable on $[a, b]$, then there exists $c \in (a, b)$ such that
$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$
$(f \in C([a, b])) \wedge (f \in C^1((a, b))) \implies \exists c \in (a, b), f'(c) = \frac{f(b)-f(a)}{b-a}$
2. Goldbach's conjecture: Every even positive integer greater than 2 can be written as the sum of two prime numbers. $\forall k \in \mathbb{N}, \exists p, q \in P, 2(k + 1) = p + q$.

Proposition 3. *Every universally quantified statement*
$$\forall x \in X, S(x)$$
can be written as a conditional statement
$$x \in X \implies S(x)$$

Example.

1. Rewrite Mean Value theorem as a universally quantified statement
2. Rewrite Goldbach conjecture as a conditional statement

Remark: Be careful with words. "At least one of the integers x and $x + 1$ is even" is not "x is even AND $x + 1$ is even".

Example. Write in symbolic logic:

1. if p is prime, then $\sqrt{p}$ is not a rational number.
2. For every positive number ε, there is a positive number M for which $|x-a| < \delta$ implies $|f(x)-f(a)| < \varepsilon$.

Negating statements

Rewrite DeMorgan's Laws, and negation of implication. $\sim (P \implies Q) = P \wedge \sim Q$.

Example.
1. If p is prime, then $\sqrt{p}$ is not rational.
2. If f is a polynomial of degree greater or equal to two, then f' is not constant.
3. $x^2 + 1 \neq 0$ for every $x \in \mathbb{R}$ (start negating quantifiers)
4. There exists a natural number n such that $2^n - 1$ is prime

Elaborate on negation of $\forall$ and $\exists$ for any statement.

- $\sim (\forall x \in X, P) = \exists x \in X, \sim P$
- $\sim (\exists x \in X, P) = \forall x \in X, \sim P$

Example. Negate the statements

1. $\forall \varepsilon > 0, \exists \delta > 0,$
$$|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$
2. $\exists M \in \mathbb{R}$, $\forall x \in [a, b]$,
$$|f(x)| \leq M$$

Workshop

Problem 1. Let X denote the set of all people, and define the following open sentences

- $S(x)$ means x is a student
- $P(x)$ means x likes pizza

Write the following statements in symbolic form.

1. All students like pizza.
2. Only students like pizza.
3. There is a student who does not like pizza.
4. Everyone either likes pizza or is a student.
5. Everyone likes pizza, or everyone is a student.

Problem 2. For each proposition, state whether it is a universal statement or an existential statement, and whether it could be verified to be true by an example.
For example, the proposition "There exists integers x and y such that $x + y = 2023$" can be verified to be true by providing an example. Then write the negation of the statement and verify if that is a universal or existential statement, and whether it could be verified to be true by an example.

a) The equation $x^3 + y^3 = 8$ has some solution, where x, y are all integers.

b) Any map on the surface of a donut can be properly colored with no more than seven colors.

c) Every mathematician knows that one cannot comb a hairy ball.

Problem 3. Explain when the given statement would be false, and write out its negation. For instance, the statement "Every person named Bernardo is Brazilian" would be false if one could find a person named Bernardo that had a different nationality.

a) If x is a multiple of 6, then x is a multiple of 3.

b) If x is prime, then x is odd.

c) If f is continuous on $[a, b]$, then it attains it's maximum value on the interval.

d) There exists irrational numbers a and b such that a^b is a rational number.

e) The decimal representation of the number π contains every possible finite sequence of digits.